

Notes: Khorana, Servaes & Wedge (JFE, 2007)

Portfolio Manager Ownership and Fund Performance

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1 Big picture

- **Question.** Does a mutual fund manager’s personal investment in the fund help predict future fund performance?
- **Regulatory setting.** The paper uses newly required SEC disclosures of portfolio-manager ownership stakes in mutual funds. Managers disclose ownership in coarse dollar ranges, not exact amounts.
- **Main empirical object.** The central variable is the portfolio managers’ aggregate dollar ownership in the fund, divided by fund assets, measured at year-end 2004.
- **Main finding.** Managerial ownership is positively related to future 2005 performance. Depending on specification, a one-basis-point increase in ownership is associated with roughly three to five basis points higher future excess performance.
- **Governance contrast.** Board characteristics—board size, board independence, and board compensation—do not robustly predict future performance once ownership and controls are included.
- **Endogeneity concern.** Ownership is not randomly assigned. Good managers may buy more shares, firms may require ownership, or skilled managers may sort into certain funds.
- **Key additional evidence.** Among managers running multiple funds, ownership is higher in the funds that subsequently perform better. This helps rule out a pure unobserved-manager-skill story.
- **Decomposition evidence.** Both predicted ownership and residual ownership are positively related to future performance. The residual component suggests that ownership disclosure contains information not already captured by observable fund, family, board, and manager variables.
- **Economic takeaway.** Manager ownership works as a useful signal for fund investors: either it improves incentives, reveals the manager’s private information about expected performance, or both.

2 Incremental contribution of the paper

- **New disclosure-based evidence.** The paper is among the first to exploit the SEC’s new portfolio-manager ownership disclosures to study whether “skin in the game” predicts mutual fund performance.
- **Bridge between mutual fund governance and corporate insider ownership.** Corporate finance has long studied insider ownership; this paper moves that logic into open-end mutual funds, where the manager is not a corporate insider but the agent responsible for portfolio decisions.
- **Predictability contribution.** The paper identifies a fund characteristic with economically meaningful predictive power, a setting where the literature had found relatively few reliable predictors of future performance.
- **Governance comparison.** It directly compares manager ownership to board-governance proxies and finds that the manager-level incentive/information variable is more important for future performance than board structure.
- **Endogeneity diagnostics.** The paper does not claim a clean causal experiment, but it adds credible diagnostics: lagged ownership, performance controls, objective fixed effects, manager fixed effects for multi-fund managers, ownership-determinant regressions, and predicted/residual ownership decomposition.
- **Investor-usefulness contribution.** The findings support the regulatory argument that ownership disclosure is useful to investors because it gives information about incentives or private expectations that is not obvious from standard fund characteristics.

3 Data and measurement

3.1 Ownership disclosure data

The disclosure rule requires managers to report holdings in ranges:

$$0, \quad \$1\text{--}\$10,000, \quad \$10,001\text{--}\$50,000, \quad \$50,001\text{--}\$100,000, \quad \$100,001\text{--}\$500,000, \\ \$500,001\text{--}\$1,000,000, \quad \text{and above } \$1,000,000.$$

The authors gather disclosures for funds filing annual reports between March and December 2005 and focus on funds with ownership data as of year-end 2004. The final performance-predictability sample contains 1,406 funds.

Interpretation: The timing is important: ownership is measured before the 2005 performance outcome. This supports a predictive interpretation, although it does not by itself prove causality.

3.2 Constructing ownership measures

The paper converts range disclosures into dollar ownership in two main ways.

- **Ownership (low):** use the bottom value of each range; this is the conservative measure.
- **Ownership (average):** use the midpoint of each range; for the above \$1 million category, use \$1 million.

For funds with multiple managers, the paper sums ownership across all managers. Managerial ownership in percent is:

$$Ownership_i = \frac{\sum_{m \in i} DollarOwnership_{mi}}{FundAssets_i} \times 100.$$

Interpretation: The percentage measure scales ownership by fund assets. This matters because a \$100,000 stake is economically different in a small fund than in a multi-billion-dollar fund.

3.3 Performance measures

The paper uses two measures of 2005 performance.

1. Objective-adjusted return (OAR):

$$OAR_i = Return_{i,2005} - MedianReturn_{objective(i),2005}.$$

2. **Four-factor alpha:** abnormal return from factor models estimated separately for equity and bond funds. For equity funds, the factors include the Fama–French factors plus momentum. For bond funds, the factors include government, corporate, mortgage-backed, long-term government, and intermediate-term government bond factors.

Interpretation: OAR is simple and objective-specific but may not fully adjust for risk. Four-factor alpha is more model-based but better controls for systematic exposures.

3.4 Governance and control variables

Governance measures include:

- board size;
- percentage of independent directors;

- board member compensation.

Fund and family controls include:

- fund size;
- family assets;
- expenses;
- front-end and back-end loads;
- portfolio turnover;
- single-manager dummy;
- investment objective dummies.

Interpretation: These controls are needed because ownership may be correlated with fund type, fund family structure, manager tenure, and other characteristics that may themselves predict performance.

4 Models and empirical specifications

4.1 Baseline performance regression

The baseline regression is:

$$Performance_{i,2005} = \alpha + \beta Ownership_{i,2004} + \gamma' X_{i,2005} + \delta_{objective(i)} + \varepsilon_i.$$

Here *Performance* is either objective-adjusted return or four-factor alpha, X_i is the vector of fund, board, family, and manager controls, and $\delta_{objective(i)}$ are objective fixed effects.

Interpretation: The coefficient β asks whether funds whose managers owned more at year-end 2004 performed better in 2005, conditional on observable characteristics.

4.2 Alternative ownership and subsample models

The paper estimates variants using:

- ownership measured using low-end range values;
- ownership measured using midpoint range values;
- average manager ownership in multi-manager funds;
- single-manager and multiple-manager subsamples;
- funds with turnover above 20%.

Interpretation: These tests check whether the ownership-performance relation is driven by the coarse range measurement, manager-team structure, or low-trading funds for which manager skill may be harder to observe.

4.3 Manager fixed-effect specification

For managers who run multiple funds, the paper estimates a model of the form:

$$Performance_{ij} = \mu_j + \beta Ownership_{ij} + \gamma' X_{ij} + \delta_{objective(i)} + \varepsilon_{ij},$$

where μ_j is a manager fixed effect.

Interpretation: This is a powerful diagnostic. It compares funds managed by the same person. If the same manager owns more of the funds that subsequently do better, the result is less likely to be driven by unobserved manager talent alone.

4.4 Determinants of ownership

The ownership-determinants regression is:

$$Ownership_{i,2004} = \alpha + \theta PastPerformance_i + \phi' Z_i + \delta_{objective(i)} + u_i.$$

Z_i includes board, fund, family, load, expense, turnover, size, and manager-tenure variables.

Interpretation: This model explains which fund characteristics are associated with manager ownership. It is not the main performance test; it is used to understand the ownership choice and to construct predicted and residual ownership.

4.5 Predicted/residual decomposition

The paper decomposes ownership into predicted and residual components:

$$Ownership_i = \widehat{Ownership_i} + \hat{u}_i.$$

Then it estimates:

$$Performance_{i,2005} = \alpha + \beta_1 \widehat{Ownership_i} + \beta_2 \hat{u}_i + \gamma' X_i + \varepsilon_i.$$

Interpretation: If only predicted ownership matters, ownership is just summarizing other observable characteristics. If residual ownership matters, the disclosure contains new information for investors.

5 Identification and empirical design

5.1 What the paper can identify

The paper is best interpreted as a performance-predictability and governance-disclosure study, not a randomized causal design. It identifies whether lagged manager ownership predicts subsequent abnormal fund performance after controlling for observable fund, family, board, and manager characteristics.

Interpretation: The causal question is difficult because managers choose whether and how much to invest. Ownership may reflect incentives, private information, or sorting. The paper's empirical design is built to separate these interpretations as much as possible using cross-sectional and within-manager evidence.

5.2 Timing and lag structure

Ownership is measured at year-end 2004. Performance is measured in 2005:

$$Ownership_{2004} \rightarrow Performance_{2005}.$$

Interpretation: This timing rules out the simplest reverse-causality problem in which 2005 performance directly determines measured ownership. However, managers may buy shares in anticipation of future performance, which is economically important but not a pure incentive effect.

5.3 Observable-control strategy

The baseline regressions include controls for fund size, family size, loads, expenses, turnover, management-team structure, objective category, and board governance.

Interpretation: The coefficient on ownership is not simply comparing equity funds with bond funds or small funds with large funds. It asks whether ownership predicts performance within a relatively comparable set of funds.

5.4 Manager fixed effects

The manager fixed-effect tests compare multiple funds run by the same manager:

$$Performance_{ij} = \mu_j + \beta Ownership_{ij} + Controls_{ij} + \varepsilon_{ij}.$$

Interpretation: This addresses a central omitted-variable concern: high-skill managers might own more and perform better. With manager fixed effects, the comparison is across funds managed by the same person. A positive coefficient means the manager owns more in the fund that later does better.

5.5 Predicted versus residual ownership

The ownership decomposition separates ownership that is predictable from observables from ownership that remains unexplained:

$$Ownership = PredictedOwnership + ResidualOwnership.$$

Interpretation: Predicted ownership may capture systematic incentive and sorting patterns. Residual ownership is more likely to capture manager-specific private information, unobserved incentives, or idiosyncratic voluntary investment. The paper finds that both matter.

5.6 Remaining identification limits

- The paper cannot distinguish cleanly between the incentive channel and the private-information channel.
- Ownership disclosures are coarse ranges, not exact dollar stakes.
- The analysis focuses on one performance year after the first disclosure wave.
- Funds may differ in unobserved ways that correlate with both ownership and future performance.
- Managers may be required by some families to hold fund shares, although the paper argues this is not common in the sample period.

Interpretation: These limits do not overturn the investor-usefulness result. Even if ownership reflects private information rather than incentives, investors can still use ownership disclosure to predict future performance.

6 Tables and figures

6.1 Table 1: Ownership of Portfolio Managers

Read:

Table 1

Ownership of portfolio managers

This table reports the dollar amount and % ownership of portfolio managers in their own funds, as of year-end 2004. All funds are classified into the following six categories: balanced, bond, equity, international bond, international equity, and sector. The mean, 50th, 75th, 90th, and 100th percentile of manager ownership figures are reported. Funds report only the range of each manager's holdings (\$0, \$1-\$10,000, \$10,001-\$50,000, \$50,001-\$100,000, \$100,001-\$500,000, \$500,001-\$1,000,000, or above \$1,000,000). In Panel A, we convert these ranges into dollars by using the lowest value of the range and sum up across all managers in a fund. In Panel B, we convert them into dollars by using the average (midpoint) of each category and sum up across all managers in a fund. For the above \$1 million category, we set the ownership level at the bottom of the range. Managerial ownership percentage is computed as the dollar ownership of all the portfolio managers of a fund, divided by fund assets. % own refers to the percentage of funds with positive ownership.

Panel A. Summary statistics of manager holdings (based on lowest value of range)

Fund type	N	% own	Managerial ownership (in dollars)					Managerial ownership (in percent)				
			Mean	50th	75th	90th	100th	Mean	50th	75th	90th	100th
All funds	1,406	43	96,663	0	50,001	160,003	3,700,006	0.04	0.00	0.01	0.09	0.98
Balanced	62	47	82,904	0	20,003	110,002	2,000,002	0.03	0.00	0.01	0.15	0.33
Domestic bond	405	26	15,444	0	1	50,001	600,002	0.01	0.00	0.00	0.02	0.98
Domestic equity	606	51	154,861	1	100,001	510,003	3,700,006	0.05	0.00	0.02	0.15	0.97
International bond	26	42	10,001	0	10,001	50,001	100,001	0.01	0.00	0.01	0.03	0.14
International equity	158	47	84,115	0	100,001	200,002	3,000,003	0.05	0.00	0.02	0.09	0.90
Sector	149	56	115,034	1	50,001	500,001	2,000,002	0.05	0.00	0.02	0.18	0.69

Panel B. Summary statistics of manager holdings (based on midpoint of range)

Fund type	N	Managerial ownership (in dollars)					Managerial ownership (in percent)				
		Mean	50th	75th	90th	100th	Mean	50th	75th	90th	100th
All funds	1,406	149,570	0	75,000	405,000	4,350,003	0.08	0.00	0.03	0.22	2.91
Balanced	62	134,274	0	65,000	330,000	2,000,002	0.08	0.00	0.02	0.33	0.93
Domestic bond	405	36,219	0	5,000	75,000	1,050,000	0.03	0.00	0.00	0.04	2.24
Domestic equity	606	226,227	5,000	300,000	885,000	4,350,003	0.11	0.00	0.06	0.35	2.91
International bond	26	25,000	0	30,000	75,000	300,000	0.03	0.00	0.03	0.08	0.42
International equity	158	161,203	0	210,000	600,000	3,000,003	0.11	0.00	0.06	0.24	2.21
Sector	149	161,913	5,000	75,000	750,000	2,000,002	0.09	0.00	0.04	0.34	1.47

Table 1: Ownership of portfolio managers. The table reports manager ownership in dollars and as a percentage of fund assets, using low-end and midpoint conversions from the SEC disclosure ranges. The central reading is that the average stake is small, but the cross-sectional distribution has enough variation to test whether ownership predicts future performance.

- The table documents how much money portfolio managers have invested in their own funds as of year-end 2004.
- Using the conservative low-end measure, 43% of funds have positive manager ownership.
- Average ownership is \$96,663 in dollar terms and 0.04% of fund assets.
- Domestic equity funds have the highest average low-end ownership: \$154,861 and 0.05% of assets.
- Sector funds have the highest fraction with positive ownership at 56%.
- Using midpoint ranges roughly doubles average ownership for all funds to \$149,570 and 0.08% of assets.

Economic message: The first empirical fact is that ownership is common but modest. This makes the later performance result striking: even small percentage stakes can carry useful incentive or information content.

How it fits the paper: The table justifies treating manager ownership as a meaningful cross-sectional variable. The variation across fund types also motivates objective fixed effects.

6.2 Table 2: Summary Statistics of Fund Characteristics in 2005

Read:

- Panel A shows that the sample funds are broadly similar to the rest of the mutual fund universe in performance and standard fund characteristics.
- The sample funds have average board size of about eight and about 77% independent directors.
- Panel B compares funds with positive manager ownership to those without ownership.
- Funds with manager ownership have higher average and median performance in 2005.
- Funds with ownership are larger, belong to smaller families, and pay higher board compensation.

Economic message: The sample does not look obviously selected on performance relative to the fund universe, but within the sample ownership funds differ from zero-ownership funds. This is why multivariate regressions are necessary.

How it fits the paper: Table 2 sets up the regression design: ownership is correlated with several fund characteristics, so the paper must control for fund size, family size, board variables, expenses, loads, and objective categories.

Table 2

Summary statistics of fund characteristics in 2005

The following variables are in percent: % independent directors, objective-adjusted return, four-factor alpha, expenses, front-end load, back-end load, and portfolio turnover. Fund size and family assets are reported in millions of dollars. Board size is the number of directors on the fund board. % independent directors is the percentage of the board members who are unaffiliated with the fund management company. Board member compensation is computed as total board compensation divided by board size. Objective-adjusted return is computed as the return on the fund less the return of the median fund with the same investment objective. Four-factor alphas are computed using separate sets of factors for equity and bonds funds. The factors are described in Section 3.2. Money market funds are excluded from the analysis. Panel A compares the sample funds with funds in the rest of the universe. Panel B compares the sample funds with fund manager ownership with the sample funds without managerial ownership. Measures of board effectiveness and the single manager dummy are gathered only for the funds in our sample, hence not available for the rest of the universe (listed as n/a). *, **, and *** indicate that the difference between the two groups being compared is significant at the 10%, 5%, and 1% levels respectively. A *t*-test is conducted for differences in means and a rank sum test for differences in medians.

Panel A. Comparing sample funds with the rest of the universe

Variable	Sample funds			Rest of universe		
	<i>N</i>	Mean	Median	<i>N</i>	Mean	Median
<i>Measures of board effectiveness</i>						
Board size	1,325	8.17	8	n/a	n/a	n/a
% independent directors	1,325	77	75	n/a	n/a	n/a
Board member compensation	1,325	1,766	748	n/a	n/a	n/a
<i>Performance measures</i>						
Return	1,327	7.29	4.85	4,952	7.11	4.92
Objective-adjusted return	1,327	0.79	0.03	4,952	0.57	-0.01
Four-factor alpha	1,255	1.16	0.01	4,614	0.98	-0.09
<i>Control variables</i>						
Single manager dummy	1,325	0.49	0	n/a	n/a	n/a
Family assets	1,325	95,029	17,794	4,947	85,220	18,110
Expenses	1,321	1.27	1.18	4,913	1.28	1.17
Fund size	1,321	1,413	271	4,832	1,622	278
Front-end load	1,327	1.53**	0*	4,952	1.37**	0*
Back-end load	1,327	0.13	0	4,952	0.14	0
Portfolio turnover	1,315	92	51**	4,857	96	55**

Panel B. Comparing funds with and funds without managerial ownership

Variable	With ownership			No ownership		
	<i>N</i>	Mean	Median	<i>N</i>	Mean	Median
<i>Measures of board effectiveness</i>						
Board size	581	8.33*	8	744	8.04*	8
% independent directors	581	77	77	744	77	75
Board member compensation	581	2,546***	1,117***	744	1,156***	503***
<i>Performance measures</i>						
Return	581	8.70***	6.39***	746	6.20***	3.99***
Objective-adjusted return	581	1.44***	0.21**	746	0.29***	-0.02**
Four-factor alpha	552	1.88***	0.45***	703	0.59***	-0.21***

(a) Panel A and beginning of Panel B.

Table 2 (continued)

Panel B. Comparing funds with and funds without managerial ownership

Variable	With ownership			No ownership		
	<i>N</i>	Mean	Median	<i>N</i>	Mean	Median
<i>Control variables</i>						
Single manager dummy	581	0.45**	0**	744	0.51**	1**
Family assets	581	84,108*	16,290*	744	103,557*	26,524*
Expenses	579	1.29	1.22...	742	1.26	1.13***
Fund size	576	1,873***	360***	745	1,057***	204***
Front-end load	581	1.44	0.00	746	1.60	0.00
Back-end load	581	0.13	0.00	746	0.14	0.00
Portfolio turnover	579	76	51	736	105	51

(b) Panel B continuation.

Table 2: Summary statistics of fund characteristics in 2005. The paired panels compare the sample to the broader universe and then compare funds with and without positive manager ownership. The pattern shows why ownership is not a random treatment: ownership funds differ in size, family affiliation, compensation, and performance.

6.3 Table 3: Explaining Fund Performance for 2005

Read:

- Panel A is the core performance regression. Ownership is positive and statistically significant across OAR and alpha specifications.
- In the most controlled OAR model, the ownership coefficient is about 3.65; in the analogous alpha model,

Table 3 (continued)

Panel C. Using manager fixed effects for domestic funds with a single manager

Variable	Objective-adjusted return		Alpha	
	(1)	(2)	(3)	(4)
Expenses		-0.78 (0.32)		0.05 (0.95)
Fund size		0.16 (0.39)		0.17 (0.40)
Back-end load		1.61 (0.54)		2.97 (0.28)
Front-end load		-0.21 (0.10)		-0.21 (0.11)
Portfolio turnover		0.23 (0.26)		0.19 (0.40)
Intercept	10.21 (0.14)	4.55 (0.59)	9.92 (0.15)	1.37 (0.87)
Objective dummies	Yes	Yes	Yes	Yes
N	554	544	532	523
Adjusted R^2	0.66	0.69	0.70	0.70

Table 3: Explaining fund performance for 2005, Panel A. The basic regressions show positive ownership coefficients for both objective-adjusted returns and four-factor alphas. The key comparison is that ownership remains significant after adding board characteristics, fund controls, and objective dummies.

it is about 3.30.

- Board variables are not robust performance predictors.
- Portfolio turnover is negatively related to performance in several specifications.
- Panel B shows the ownership result survives alternative ownership definitions and subsamples.
- Panel C uses manager fixed effects for domestic single-manager funds and continues to find positive ownership-performance coefficients.

Economic message: Table 3 is the main predictive-performance table. It shows that a higher manager stake predicts better subsequent performance even after controlling for governance, fund, family, and objective characteristics.

How it fits the paper: The manager fixed-effect panel is especially important. It shows that the result is not only that better managers own more, but that within a manager's own menu of funds, larger ownership is associated with better future performance.

Table 3 (continued)

Panel B. Alternative models using objective-adjusted returns

Variable	Full sample	Single manager	Multiple managers	Full sample	Turnover > 20%
	(1)	(2)	(3)	(4)	(5)
Ownership (low)		4.30 (0.00)	2.93 (0.03)		4.36 (0.00)
Ownership (average)	1.49 (0.00)				
Average manager ownership				4.56 (0.00)	
Board size	0.05 (0.30)	0.02 (0.73)	0.08 (0.21)	0.05 (0.31)	0.06 (0.28)
% independent directors	-0.01 (0.53)	-0.01 (0.68)	-0.01 (0.37)	-0.01 (0.50)	-0.01 (0.38)
Board member compensation	-0.06 (0.18)	-0.11 (0.12)	-0.05 (0.46)	-0.06 (0.20)	-0.04 (0.45)
Family assets	-0.01 (0.92)	-0.09 (0.37)	0.09 (0.41)	-0.00 (0.98)	-0.02 (0.81)
Single manager dummy	-0.03 (0.89)			-0.13 (0.56)	-0.04 (0.89)
Expenses	0.25 (0.21)	-0.37 (0.32)	0.45 (0.07)	0.25 (0.21)	0.42 (0.06)
Fund size	0.45 (0.00)	0.41 (0.00)	0.41 (0.00)	0.44 (0.00)	0.48 (0.00)
Back-end load	-0.11 (0.46)	0.02 (0.95)	-0.17 (0.37)	-0.12 (0.44)	-0.17 (0.32)
Front-end load	-0.06 (0.25)	0.02 (0.73)	-0.14 (0.07)	-0.06 (0.28)	-0.06 (0.28)
Portfolio turnover	-0.17 (0.03)	0.08 (0.46)	-0.36 (0.00)	-0.18 (0.02)	-0.23 (0.01)
Intercept	-2.75 (0.03)	-1.47 (0.40)	-3.03 (0.10)	-2.68 (0.03)	-2.68 (0.05)
Objective dummies	Yes	Yes	Yes	Yes	Yes
N	1,243	600	643	1,243	983
Adjusted R^2	0.04	0.03	0.05	0.04	0.05

Panel C. Using manager fixed effects for domestic funds with a single manager

Variable	Objective-adjusted return		Alpha	
	(1)	(2)	(3)	(4)
Ownership (low)	5.04 (0.03)	4.15 (0.06)	4.60 (0.07)	4.53 (0.08)
Board size	0.10 (0.73)	-0.03 (0.93)	-0.04 (0.88)	-0.28 (0.40)
% independent directors	-0.21 (0.08)	-0.14 (0.41)	-0.14 (0.21)	0.02 (0.89)
Board member compensation	0.02 (0.86)	0.05 (0.69)	0.13 (0.21)	0.07 (0.56)
Family assets	0.48 (0.24)	0.35 (0.51)	0.27 (0.50)	-0.14 (0.80)

(a) Panel B and start of Panel C.

(b) Panel C continuation.

Table 3: Explaining fund performance for 2005, Panels B–C. Panel B demonstrates robustness to ownership measurement and manager-team structure. Panel C adds manager fixed effects, reducing concern that unobserved manager skill alone drives the result.

Table 3 (continued)

Panel C. Using manager fixed effects for domestic funds with a single manager

Variable	Objective-adjusted return		Alpha	
	(1)	(2)	(3)	(4)
Expenses		-0.78 (0.32)		0.05 (0.95)
Fund size		0.16 (0.39)		0.17 (0.40)
Back-end load		1.61 (0.54)		2.97 (0.28)
Front-end load		-0.21 (0.10)		-0.21 (0.11)
Portfolio turnover		0.23 (0.26)		0.19 (0.40)
Intercept	10.21 (0.14)	4.55 (0.59)	9.92 (0.15)	1.37 (0.87)
Objective dummies	Yes	Yes	Yes	Yes
N	554	544	532	523
Adjusted R^2	0.66	0.69	0.70	0.70

6.4 Table 4: Determinants of Fund Manager Ownership at Year-End 2004

Read:

- Ownership is higher in funds with better past objective-adjusted performance and better past alpha.
- Ownership is lower in larger funds and funds belonging to larger families.
- Front-end loads are negatively related to ownership, suggesting managers invest less in funds with investor-facing sales loads.
- Average manager tenure is positively related to ownership.
- Equity, sector, and international equity funds have higher ownership than the omitted international-bond category.
- Board member compensation is positively associated with ownership, but board size and independence are generally not important determinants.

Economic message: Table 4 shows that ownership is endogenous and predictable. Managers tend to own more where past performance is better, tenure is longer, funds are smaller, and asset classes are equity-oriented.

How it fits the paper: This table is the foundation for Table 5. By modeling ownership, the authors can ask whether the predictable and residual pieces of ownership each predict future performance.

Table 4

Determinants of fund manager ownership at year-end 2004

This table reports ordinary least squares (OLS) and interval regression results using fund manager ownership in percent at the end of 2004 as the dependent variable. The OLS models (Models (1)-(4)) use fund manager ownership based on the lowest value of the ownership range. The interval regression model (Model (5)) uses both the lowest and highest possible levels of fund manager ownership in each range. All explanatory variables are measured in 2004 unless indicated otherwise. The objective-adjusted return (OAR) is computed as the annual return of the fund less the return of the median fund in the matched investment objective. Alpha is the abnormal fund return estimated using separate four-factor models for equity and bond funds over the period 2002-2004. Volatility is the annualized standard deviation of the monthly returns in 2003 and 2004. Board size is the number of directors on the fund board. % independent directors is the percentage of the board members who are unaffiliated with the fund management company. Board member compensation is the log of total board compensation divided by board size. Family assets is the log of total assets of the fund management company. Single manager dummy is equal to one if the fund is managed by one manager and zero otherwise. Fund size is measured as the log of total assets. Average tenure is the average tenure of all the fund managers managing a particular fund. A dummy variable is included for each investment objective, with the exception of international bond funds. Expenses and loads are measured in percent. Models (3)-(5) are limited to the sample with average manager tenure longer than one year at the end of 2005. The numbers reported in parentheses are *P*-values. All coefficients are multiplied by one hundred.

Variable	Ordinary least squares regression				Interval regression
	Full sample (1)	Full sample (2)	Average tenure > 1 yr (3)	Average tenure > 1 yr (4)	Average tenure > 1 yr (5)
OAR 2004	0.17 (0.01)		0.22 (0.01)		0.24 (0.01)
OAR 2003	0.09 (0.00)		0.12 (0.01)		0.15 (0.00)
Alpha 2002-2004		0.33 (0.00)		0.35 (0.00)	
Volatility 2003-2004	-0.59 (0.12)	0.13 (0.73)	-0.70 (0.11)	0.21 (0.63)	-0.79 (0.13)
Board size	-0.15 (0.31)	-0.08 (0.62)	-0.08 (0.66)	0.02 (0.91)	-0.08 (0.68)
% independent directors	-0.03 (0.36)	-0.03 (0.40)	0.04 (0.34)	0.03 (0.36)	0.03 (0.41)
Board member compensation	0.45 (0.01)	0.51 (0.01)	0.37 (0.08)	0.40 (0.07)	0.51 (0.03)
Family assets	-0.56 (0.03)	-0.57 (0.04)	-0.37 (0.19)	-0.43 (0.14)	-0.53 (0.10)
Single manager dummy	-0.61 (0.41)	-0.78 (0.30)	-1.06 (0.22)	-1.46 (0.11)	-1.43 (0.15)
Expenses	1.27 (0.18)	1.12 (0.26)	0.94 (0.45)	0.85 (0.52)	1.52 (0.26)
Fund size	-0.82 (0.00)	-0.90 (0.00)	-1.57 (0.00)	-1.58 (0.00)	-1.50 (0.00)
Back-end load	0.17 (0.63)	0.38 (0.32)	0.17 (0.67)	0.45 (0.30)	-0.00 (0.99)
Front-end load	-0.41 (0.08)	-0.40 (0.11)	-0.58 (0.03)	-0.60 (0.03)	-0.89 (0.00)
Average tenure			0.62 (0.00)	0.64 (0.00)	0.82 (0.00)

Table 4: Determinants of fund manager ownership at year-end 2004, first page. The first page reports OLS and interval-regression models linking ownership to past performance, volatility, board variables, size, loads, and tenure. Past performance and smaller fund size are central predictors.

Table 4 (continued)

Variable	Ordinary least squares regression				Interval regression
	Full sample (1)	Full sample (2)	Average tenure > 1 yr (3)	Average tenure > 1 yr (4)	Average tenure > 1 yr (5)
Equity dummy	5.65 (0.00)	6.64 (0.00)	7.84 (0.00)	8.45 (0.00)	9.92 (0.00)
Sector dummy	5.07 (0.01)	4.59 (0.02)	6.02 (0.01)	5.09 (0.02)	7.63 (0.01)
Intl equity dummy	3.75 (0.03)	3.23 (0.06)	4.50 (0.03)	3.57 (0.08)	5.45 (0.02)
Balanced dummy	2.42 (0.06)		2.72 (0.08)		3.70 (0.07)
Bond dummy	0.48 (0.64)	2.26 (0.05)	-0.26 (0.83)	1.71 (0.21)	-0.73 (0.63)
Intercept	11.44 (0.00)	8.55 (0.01)	6.36 (0.10)	3.21 (0.44)	6.95 (0.11)
<i>N</i>	1,328	1,272	983	943	983
Adjusted <i>R</i> ²	0.10	0.09	0.16	0.15	N/A

Table 4: Determinants of fund manager ownership at year-end 2004, continuation. The continuation shows asset-class dummies. Equity-oriented funds have substantially higher ownership, implying that the economic importance of ownership is larger in funds where managers hold more personal capital.

6.5 Table 5: Explaining 2005 Performance with Decomposed Managerial Ownership

Read:

- Predicted ownership is positive and significant across models.
- Residual ownership is also positive and significant or near significant depending on the performance measure.
- The paper reports that moving predicted ownership from the 25th to 75th percentile raises performance by about 58 basis points in the relevant model.
- Moving residual ownership from the 25th to 75th percentile raises excess performance by about 21 basis points.

Table 5

Explaining 2005 performance with decomposed managerial ownership

This table reports ordinary least squares regression results in which fund performance [measured using objective-adjusted returns (OARs) and four-factor alphas] is the dependent variable. The objective-adjusted return is computed as the annual return of the fund less the return of the median fund in the matched investment objective. Alpha is the abnormal fund return estimated using separate four-factor models for equity and bond funds. The predicted and residual ownership levels are computed using equivalent models in Table 4. Fund size is the log of total net assets. Family assets is the log of total assets of the fund management company. Models (3) and (4) are limited to the sample with average manager tenure longer than one year at the end of 2005. All the control variables are measured in 2005. Numbers reported in parentheses are *P*-values.

Variable	OAR full sample	Alpha	OAR tenure > 1 yr	Alpha tenure > 1 yr
	(1)	(2)	(3)	(4)
Predicted ownership	48.82 (0.00)	18.77 (0.00)	25.90 (0.00)	7.68 (0.06)
Residual ownership	1.68 (0.07)	2.79 (0.00)	1.73 (0.10)	2.90 (0.01)
Expenses	-0.14 (0.56)	0.98 (0.00)	0.20 (0.47)	1.18 (0.00)
Fund size	0.51 (0.00)	0.56 (0.00)	0.51 (0.00)	0.56 (0.00)
Family assets	0.42 (0.00)	0.20 (0.01)	0.27 (0.00)	0.11 (0.20)
Back-end load	-0.05 (0.73)	-0.29 (0.07)	-0.04 (0.79)	-0.29 (0.08)
Front-end load	0.08 (0.10)	-0.03 (0.60)	0.05 (0.33)	-0.05 (0.37)
Portfolio turnover	-0.02 (0.75)	-0.12 (0.13)	-0.06 (0.57)	-0.19 (0.08)
Intercept	-8.64 (0.00)	-5.38 (0.00)	-7.12 (0.00)	-4.39 (0.00)
Objective dummies	Yes	Yes	Yes	Yes
<i>N</i>	1,184	1,134	925	885
Adjusted R^2	0.10	0.17	0.08	0.13

Table 5: Explaining 2005 performance with decomposed managerial ownership. This table decomposes ownership into a fitted component from Table 4 and a residual component. Both components help predict future performance, which supports the claim that manager ownership disclosure is useful beyond standard observable variables.

Economic message: Table 5 is the strongest investor-usefulness table. If residual ownership predicts future performance, the disclosure contains information not already summarized by observable fund characteristics.

How it fits the paper: The table helps separate two interpretations. Predicted ownership may reflect systematic incentives and fund characteristics. Residual ownership points toward private information or unobserved incentives. Both matter for future performance.

7 Short summary

- The paper studies whether mutual fund managers who personally own more of their funds perform better subsequently.
- It uses newly required SEC disclosures of manager ownership ranges.
- The final sample contains 1,406 funds with year-end 2004 ownership data.
- Average ownership is modest: about 0.04% of fund assets using the conservative low-end conversion.
- Domestic equity and sector funds have higher ownership than bond funds.
- Performance is measured using 2005 objective-adjusted returns and four-factor alphas.
- Manager ownership predicts future performance across baseline and robustness specifications.
- Board-governance variables do not robustly predict future performance.
- Manager fixed-effect tests show that managers owning multiple funds tend to own more of the funds that subsequently do better.
- Ownership is itself predictable from past performance, size, family size, loads, tenure, and asset class.

- Decomposing ownership into predicted and residual components shows that both components forecast performance.
- The paper cannot fully distinguish incentive alignment from private information, but both interpretations make ownership disclosure useful to investors.

8 Conclusion

8.1 Core conclusion

Khorana, Servaes, and Wedge show that portfolio-manager ownership is positively related to future mutual fund performance. Even though the average stake is modest, the cross-sectional variation in manager ownership contains economically useful information for investors.

8.2 Economic intuition

Manager ownership can matter for two reasons. First, it can improve incentives: managers bear more of the downside and enjoy more of the upside from performance. Second, it can reveal private information: managers may buy more shares in funds they expect to perform well. The paper cannot cleanly separate these two mechanisms, but from an investor’s perspective both channels make the disclosure valuable.

8.3 Policy implication

The evidence supports the SEC’s decision to require disclosure of portfolio-manager ownership. The disclosure is not merely a privacy-sensitive detail about a manager’s personal portfolio; it is a variable that predicts subsequent fund performance and is not fully subsumed by standard fund characteristics.

8.4 Relation to fund governance

The paper also suggests that manager-level incentives are closer to the performance-production function than board characteristics. Mutual fund boards may be important for fees, mergers, oversight, and compliance, but the manager’s own capital at risk is more directly linked to portfolio outcomes.

8.5 Limitations

The design is predictive rather than experimental. Ownership is chosen, disclosed in ranges, and observed during one early disclosure wave. The estimated relation may combine incentive effects, private information, family policies, and unobserved fund characteristics. These limitations explain why the paper carefully uses controls, manager fixed effects, ownership determinants, and residual decomposition rather than claiming a clean causal effect.

8.6 Takeaway

The main takeaway is that “skin in the game” matters in delegated portfolio management. A manager’s ownership stake is a compact signal of either incentive alignment or informed confidence. For investors choosing mutual funds, this information is useful because it predicts future risk-adjusted performance after controlling for many standard fund and governance variables.